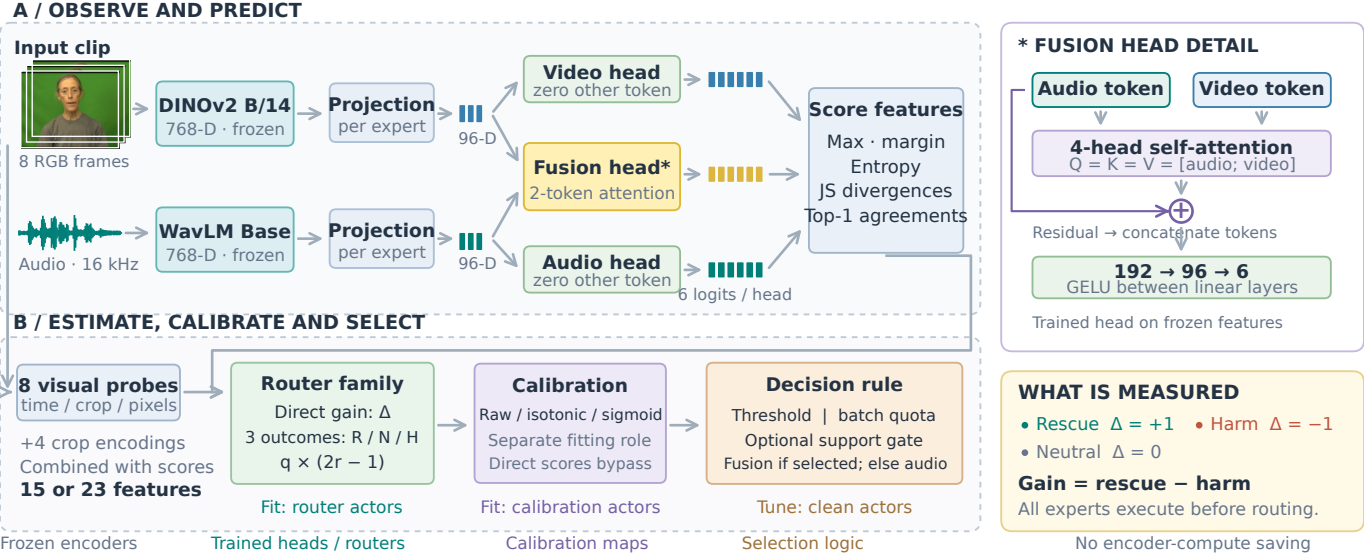

A / OBSERVE AND PREDICT
Input clip
8 RGB frames
DINOv2 B/14
768-D · frozen
Projection
per expert
96-D
Video head
zero other token
Fusion head*
2-token attention
Audio head
zero other token
Audio · 16 kHz
WavLM Base
768-D · frozen
Projection
per expert
96-D
6 logits / head
Score features
Max · margin
Entropy
JS divergences
Top-1 agreements
B / ESTIMATE, CALIBRATE AND SELECT
8 visual probes
time / crop / pixels
+4 crop encodings
Combined with scores
15 or 23 features
Router family
Direct gain: Δ
3 outcomes: R / N / H
q × (2r − 1)
Calibration
Raw / isotonic / sigmoid
Separate fitting role
Direct scores bypass
Decision rule
Threshold | batch quota
Optional support gate
Fusion if selected; else audio
Fit: router actors
Fit: calibration actors
Tune: clean actors
Frozen encoders
Trained heads / routers
Calibration maps
Selection logic
* FUSION HEAD DETAIL
Audio token
Video token
4-head self-attention
Q = K = V = [audio; video]
Residual → concatenate tokens
192 → 96 → 6
GELU between linear layers
Trained head on frozen features
WHAT IS MEASURED
• Rescue Δ = +1
• Harm Δ = −1
• Neutral Δ = 0
Gain = rescue − harm
All experts execute before routing.
No encoder-compute saving